

Tool-Call Interception: Building API-Level Guardrails for LLM Agents

Seminar Abstract

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Abstract

Autonomous AI agents built on frameworks like LangChain and AutoGen are increasingly empowered to interact with complex external environments through function calling and tool use. While these capabilities enhance operational utility, the non-deterministic nature of large language models (LLMs) introduces severe security vulnerabilities. These agents are highly susceptible to prompt injection, reasoning errors, and unintended actions that can result in unauthorized data access or complete system compromise. Current industry practices often rely on static container isolation (e.g., Docker) configured only at deployment time, which fails to adapt dynamically while the agent is running. Furthermore, traditional in-process enforcement methods, such as command denylists, have proven empirically fragile and can be easily bypassed by agents utilizing dynamic-linker tricks or shell metacharacters.

This seminar explores the mechanism of API-level tool-call interception as a robust, dynamic defense strategy, focusing on the framework proposed in the ICSE 2026 paper, *AgentSpec*. The proposed system introduces customizable runtime enforcement that acts as a software-level middleware layer. By intercepting every action an agent attempts directly at the framework boundary, the proxy evaluates requested operations against explicit, capability-based policies before any execution is permitted. This approach deterministically allows, denies, or restricts actions in real-time, ensuring agents operate strictly within designated safety boundaries. The presentation will cover the architectural integration of these API guardrails, their measurable effectiveness in reducing adversarial attack success rates, and their critical role as a reactive policy layer situated on top of foundational OS-level isolation.

Key Reference Papers

1. “AgentSpec: Customizable Runtime Enforcement for Safe and Reliable LLM Agents,” International Conference on Software Engineering (ICSE), 2026.

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